

Homework 5

Nearly Free electrons in 2d: mind the gap.

1. In this problem we will explore a periodic potential for the 2d square lattice. For your convenience, I have plotted the empty lattice below, and you will need to figure out how this is perturbed by forming the Hamiltonian matrix and diagonalizing it. Let us define the potential:

$$V(\vec{r}) = \sum_{m_1, m_2} V_{(m_1, m_2)} e^{i(m_1 \vec{b}_1 \cdot \vec{r} + m_2 \vec{b}_2 \cdot \vec{r})}$$

Let us assume a simple form with the only non-zero coefficients of:

$$V_{(1,0)} = V_{(\bar{1},0)} = V_{(0,1)} = V_{(0,\bar{1})} = -0.04 \quad V_{(1,1)} = V_{(\bar{1},\bar{1})} = V_{(\bar{1},1)} V_{(1,\bar{1})} = -0.02$$

The above potentials have units of $\frac{\hbar^2}{2m} \frac{4\pi^2}{a^2}$. In this problem, we will truncate and only keep the following nine $\vec{G}_{(m_1, m_2)}$ in our basis:

$$(0, 0), (1, 0), (\bar{1}, 0), (0, 1), (0, \bar{1}), (1, 1), (\bar{1}, \bar{1}), (1, \bar{1}), (\bar{1}, 1)$$

- (a) Make a real space plot of this potential for two separate directions: $[1,0]$ and $[1,1]$
- (b) Please write out two 9x9 matrices for the Hamiltonian in this basis. Please have one matrix for the kinetic energy, and another for the potential.
- (c) Numerically diagonalize this 9x9 matrix and plot the Eigenvalues along the path illustrated below. Please also plot the free electrons for comparison. Please plot in the same range as I did below: energies of 0-2.
- (d) Consider the k-point $(\frac{1}{2}, 0)$ and it's lowest Eigenvalue which is degenerate at the free level. Write and analytically solve the 2x2 matrix which contains the direct coupling between the two states. Add the resulting two values on your plot as large points.
- (e) Please evaluate whether or not there is any degeneracy remaining (ie. band that have identical values) once you turn on the potential. How many degenerate points are there between energies of 0-0.75? What is the size of the degeneracy (ie. two-fold, three-fold, etc)?

Free Dispersion for square lattice

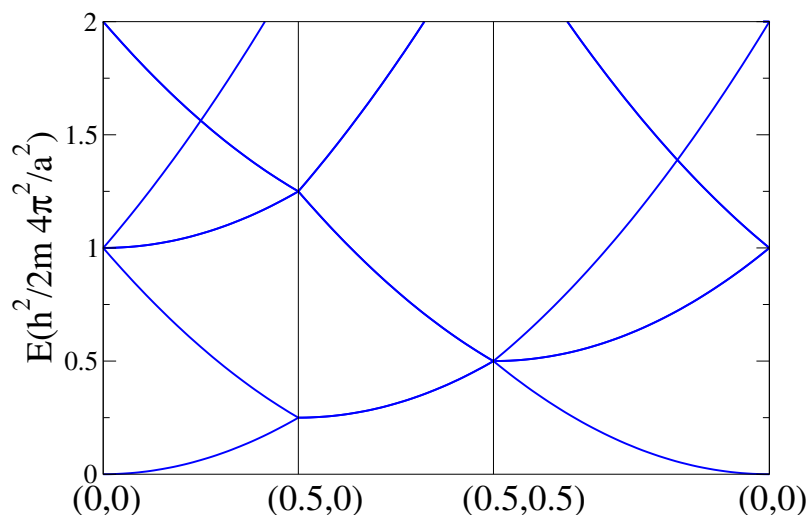


Figure 1: Free electrons for the 2d empty square lattice with parameter a . The k-point path is indicated in the x-axis in lattice coordinates (ie. fractions of the reciprocal lattice vectors $\vec{b}_1$ and $\vec{b}_2$).